

DeskFanArray

A 3D-printed bar of six PWM case fans, driven by an ESP32-S3 with a built-in touch display. Pick an airflow pattern on the screen, such as a gentle breeze, a travelling wave or a pulse, and the fans do the rest.



The finished build: six 12 V fans in a printed housing, with the controller display on a stalk in the middle.

Overview

DeskFanArray combines six individually controlled fans in one long housing that sits on a desk. A small touch screen on a stalk lets you switch between control modes and adjust their parameters with sliders and a 2D pad. Each fan gets its own PWM signal, so patterns can move across the array instead of simply running all fans at the same speed.

- **Six 80 mm fans** with independent PWM speed control (any PWM fan works).
- **Touch UI** on a Waveshare ESP32-S3 1.47" display board, with seven screens, one per mode.
- **Fully printed housing**, designed in Fusion 360.
- **Single 12 V input** (match your fans' voltage); a buck converter makes the 5 V for the controller.
- **Arduino firmware** (.ino) with a configurable startup screen.

Control UI

The display shows seven screens, one per mode. The dots at the bottom show where you are. The six grey bars mirror the current speed of each fan. The red and blue controls set the parameters of the active mode.



breeze: two sliders shape the airflow (red and blue).



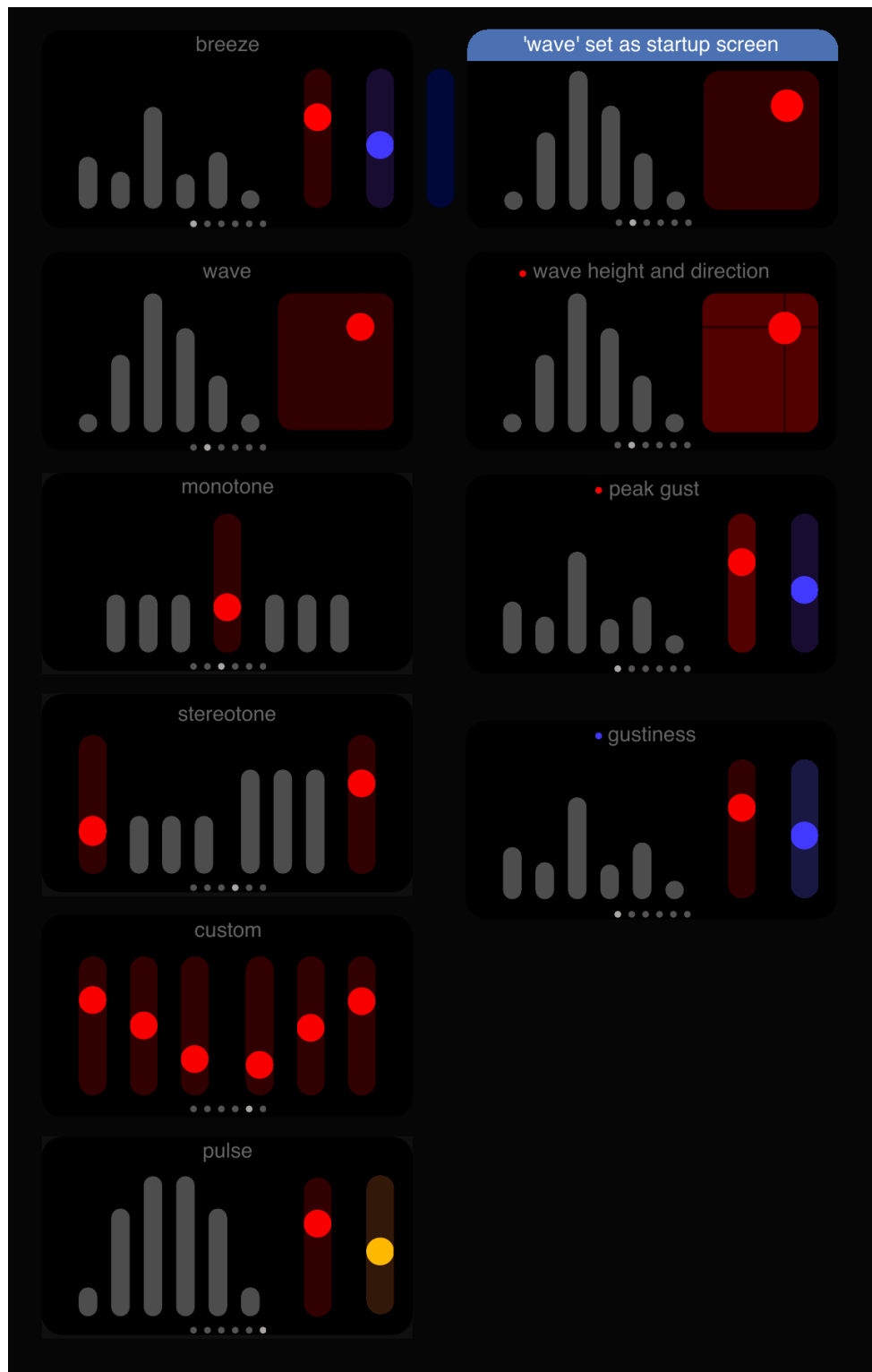
wave: a 2D pad sets wave height and direction.

Screen	Controls (from the UI design)
breeze	Two sliders: <i>peak gust</i> (red) and <i>gustiness</i> (blue)
wave	2D pad for <i>wave height</i> and <i>direction</i>
monotone	One slider sets all fans to the same speed
stereotone	Two sliders, one for each half of the array (left three, right three fans)
custom	Six sliders, one per fan
pulse	Two sliders (red and amber) for the pulsing pattern

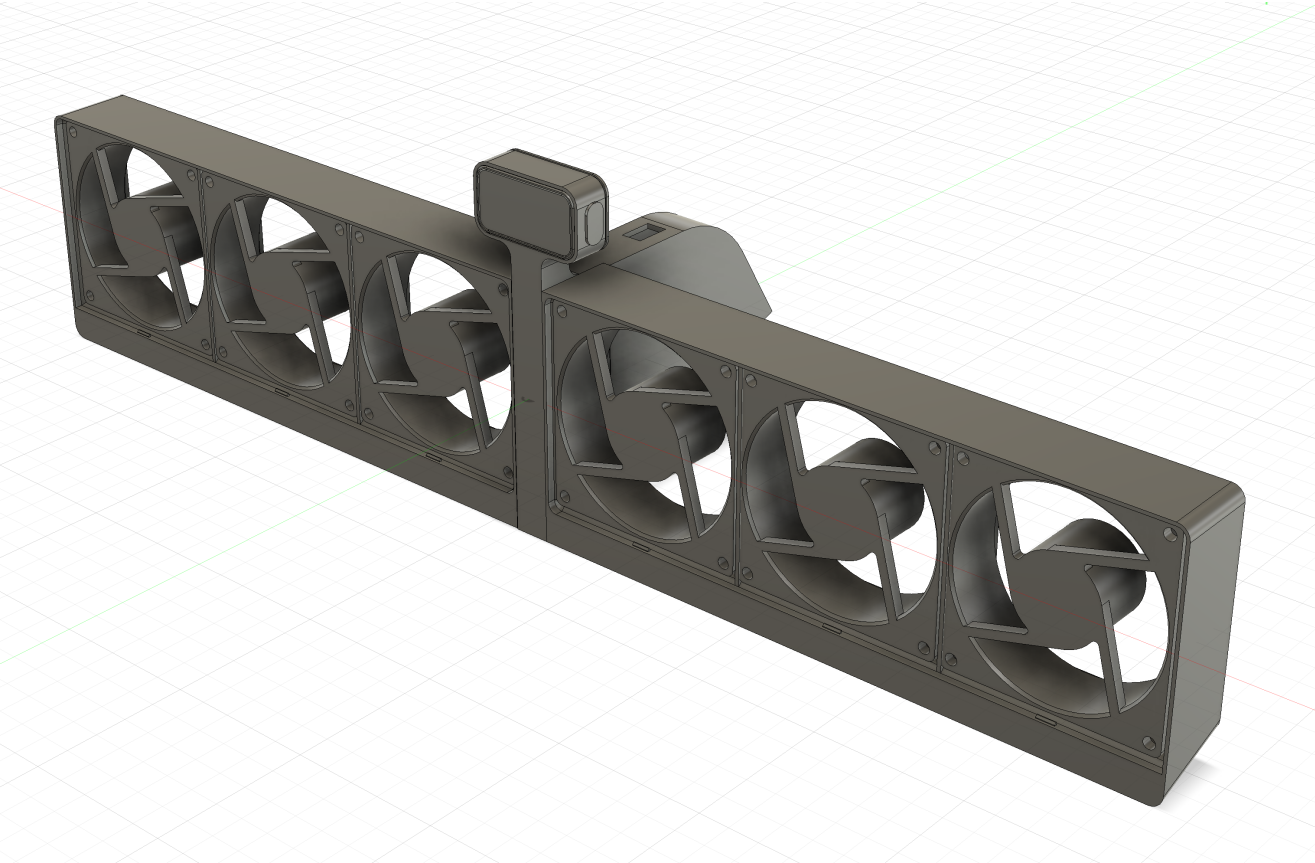
Setting the default screen: press and hold (LongPress) on a control screen to make it the default. On power-up, that control screen is shown first. The design on the next page shows wave set as the startup screen.

UI design overview

Design sketch of all screens (Sketch file in docs/). Left column: breeze, wave, monotone, stereotone, custom, pulse. Right column: the wave startup-screen setting and the sub-screens for wave height/direction, peak gust and gustiness.



Hardware



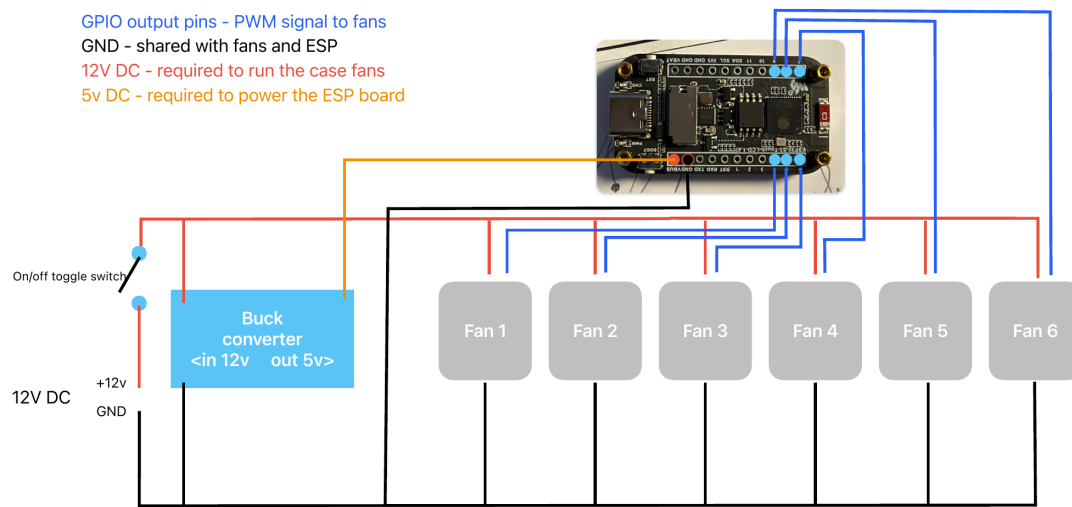
CAD model of the housing in Fusion 360 (file in models/Fusion/).

Part	Notes
ESP32-S3 touch LCD, 1.47"	Waveshare board; runs the UI and generates the six PWM signals
6 x 80x80 mm PWM case fans	Xilence XPF80.R PWM in this build (12 V). Each fan's PWM wire goes to its own GPIO pin
Buck converter	12 V in, 5 V out; powers the ESP board
12 V DC supply	Powers the fans directly and the controller via the buck converter
On/off toggle switch	Mounted on the case behind the display; switches the +12 V line to both the buck converter and the fans
3D-printed housing	Printed parts from the Fusion 360 model: two case halves and a midsection
4 x M3 bolts + nuts	Mount the two case halves to the midsection
4 x M2 bolts	Mount the controller to the midsection

About the fans. The fans are 80x80 mm. The ones used in this build are inexpensive Xilence XPF80.R PWM case fans, but any fan will do as long as it accepts a PWM input. If your fans need a voltage other than 12 V, adapt the supply voltage accordingly (and make sure the buck converter accepts that input voltage and still delivers 5 V to the controller).

Wiring

The 12 V supply passes through the on/off toggle switch, which feeds the 12 V rail for the six fans and the buck converter. All fans share a common ground with the ESP board. The buck converter drops 12 V to 5 V for the controller, and six GPIO pins send PWM signals to the fans.



IMPORTANT
Only connect the USB C plug on the ESP board when the 12V supply source is DISCONNECTED and/or main switch is OFF. Otherwise you'll have a conflicting 5V from USB and a 5V from the buck converter to the ESP controller.

Wiring diagram (PDF in docs/). Blue: PWM signals. Black: ground. Red: 12 V. Orange: 5 V.

IMPORTANT

Only connect the USB-C plug on the ESP board when the 12 V supply is disconnected and/or the main switch is OFF. Otherwise the board receives a conflicting 5 V from USB and from the buck converter.

Repository layout

```
DeskFanArray/  
  docs/      wiring diagram, UI design (Sketch), board photo, notes  
  models/    Fusion 360 housing model  
  src/DeskFanController/  
    DeskFanController.ino  firmware  
    startup_image.h        startup image data
```

Build and flash

- Open `src/DeskFanController/DeskFanController.ino` in the Arduino IDE.
- Select the Waveshare ESP32-S3 touch LCD 1.47" board (ESP32-S3 core) and the matching USB port.
- Disconnect the 12 V supply (see the warning above), connect USB-C, and upload.
- Disconnect USB, then power the unit from the 12 V supply.

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